

# Requirement Analysis Document

## Project Title

AI-Powered Document Classification System

## 1. Project Overview

The objective of this project is to design and develop an AI-powered document classification system capable of identifying and classifying uploaded documents into predefined categories such as Aadhaar Card, PAN Card, and Marksheet. The system will use Machine Learning techniques to analyze uploaded documents and provide classification results along with confidence scores.

The application will consist of a FastAPI-based backend, a Machine Learning model for document classification, and a user-friendly frontend interface for document upload and result visualization.

---

## 2. Problem Statement

Organizations and institutions often receive large numbers of documents that need to be categorized manually. Manual classification is time-consuming, error-prone, and inefficient.

This project aims to automate document classification using Artificial Intelligence, reducing manual effort and improving accuracy and processing speed.

---

## 3. Project Objectives

- Develop an AI model to classify documents.
  - Identify Aadhaar Cards, PAN Cards, and Marksheets.
  - Generate confidence scores for predictions.
  - Create REST APIs using FastAPI.
  - Build a user-friendly interface for document upload.
  - Store uploaded document information and prediction history.
  - Provide model information and system health monitoring APIs.
- 

## 4. Scope of the Project

### In Scope

- Upload document images or scanned documents.
- Classify documents into predefined categories.

- Display confidence score.
- Maintain prediction records.
- Provide API documentation through Swagger UI.
- Support future extension for additional document types.

### **Out of Scope**

- OCR-based data extraction.
  - Editing uploaded documents.
  - User role management beyond basic authentication.
  - Multi-language document support.
- 

## **5. Functional Requirements**

### **FR-1: User Authentication**

- Users should be able to register and log in.
- JWT authentication should be implemented.

### **FR-2: Document Upload**

- Users should be able to upload document images.
- Supported formats:
  - JPG
  - JPEG
  - PNG
  - PDF (optional)

### **FR-3: Document Classification**

- System should classify uploaded documents into:
  - Aadhaar Card
  - PAN Card
  - Marksheet

### **FR-4: Confidence Score**

- System should display confidence score for each prediction.

### **FR-5: Prediction History**

- Store uploaded document details and prediction results.

### **FR-6: Model Information**

- Display model name, version, and accuracy.

## **FR-7: Health Check**

- Provide endpoint to verify backend availability.
- 

## **6. Non-Functional Requirements**

### **Performance**

- Prediction response time should be less than 3 seconds.

### **Scalability**

- System should support future document categories.

### **Reliability**

- APIs should provide consistent responses.

### **Security**

- JWT-based authentication.
- Input validation.
- Secure file uploads.

### **Usability**

- Easy-to-use user interface.
  - Responsive design.
- 

## **7. Technology Stack**

### **Frontend**

- ReactJS
- HTML
- CSS
- JavaScript

### **Backend**

- FastAPI
- Python

### **Machine Learning**

- TensorFlow / PyTorch

- OpenCV
- NumPy

## Database

- PostgreSQL or SQLite

## API Documentation

- Swagger UI (FastAPI Built-in)

## Version Control

- Git
  - GitHub
- 

# 8. System Architecture

User → Frontend (React) → FastAPI Backend → ML Model → Database

Workflow:

1. User uploads document.
  2. Frontend sends file to backend.
  3. Backend preprocesses document.
  4. ML model predicts document type.
  5. Confidence score is generated.
  6. Result is stored in database.
  7. Response is displayed to user.
- 

# 9. Assumptions

- Users upload valid document images.
  - Training dataset contains sufficient samples of Aadhaar Cards, PAN Cards, and Marksheets.
  - Internet connection is available for accessing the application.
- 

# 10. Expected Outcome

The system should successfully classify uploaded documents into their respective categories with high accuracy and provide confidence scores through a secure and user-friendly web application.